

ACUTE NEUROVASCULAR PROTOCOL

CLINICAL STANDARD OPERATING PROCEDURE: ACUTE ISCHEMIC AND HEMORRHAGIC STROKE MANAGEMENT

*Hyperacute Hyperperfusion Metrics, Thrombolytic Gateways via tPA/TNK, and
Comprehensive Stroke Center Routing Blueprints*

This clinical directive formalizes the hyperacute operational frameworks governing cerebrovascular accidents at St. Jude General Hospital. It details the step-by-step timeline targets from arrival to imaging, thrombolytic calculation boundaries, endovascular thrombectomy (EVT) routing, and post-stabilization neuro-ICU care rules. Optimized for structured technical parsing and natural language querying inside the enterprise AI Knowledge Ingestion Matrix.

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1. Introduction, Critical Objectives, and 'Time is Brain' Axiom

Acute cerebrovascular accidents (CVAs) are emergency conditions where the speed of clinical intervention directly dictates long-term functional recovery and mortality outcomes. Ischemic stroke pathology involves a core area of irreversible tissue necrosis surrounded by an ischemic penumbra—a zone of hypoperfused but salvageable brain tissue that decays at an estimated rate of 1.9 million neurons per minute if blood flow is not restored.

The objective of this Standard Operating Procedure is to standardize hyperacute neurovascular operations across St. Jude General Hospital. The protocol enforces tight, time-locked targets designed to optimize door-to-needle (DTN) times for chemical thrombolysis and door-to-puncture (DTP) times for endovascular interventions, minimizing operational variance across shifts.

1.1 Global Care Metrics and Operational Targets

- **Door-to-Stroke-Team Activation:** ≤ 10 minutes from presentation.
- **Door-to-CT Scan Initiation:** ≤ 20 minutes from presentation.
- **Door-to-CT Interpretation:** ≤ 45 minutes from presentation.
- **Door-to-Needle (DTN) Time:** ≤ 45 minutes benchmark (Target ideal ≤ 30 minutes).
- **Door-to-Groin Puncture (Endovascular):** ≤ 90 minutes from presentation.

2. Departmental Responsibility Matrix and Stroke Code Ownership

Executing a hyperacute stroke code requires simultaneous, parallel workflows rather than sequential steps. The responsibilities are allocated across the following key roles:

- **Triage Nurse / Emergency Department Clerk:** Responsible for immediate identification of neurological signs using the BE-FAST tool, executing a priority check-in, logging the exact Last Known Well (LKW) time stamp, and activating the "Code Stroke" alert system.
- **Stroke Neurology Fellow / Attending Neurologist:** Accountable for executing the formal National Institutes of Health Stroke Scale (NIHSS) assessment, coordinating with radiology for immediate priority scanning, verifying inclusion/exclusion metrics for thrombolysis, and ordering the specific dosing regimen.
- **Neuro-Radiology Technologist & Radiologist:** Tasked with clearing the CT/MRI suite immediately upon a Code Stroke page, conducting unenhanced head CT and CT Angiography (CTA) maps, and providing an immediate verbal read for intracranial hemorrhage checks.
- **ED Nursing Staff:** Accountable for securing dual large-bore IV lines (18g preferred), drawing baseline STAT labs (including point-of-care glucose and coagulation profiles), managing active blood pressure parameters, and mixing thrombolytic agents at the bedside.

3. Pre-Hospital Routing & Hyperacute Emergency Triage Framework

The stroke code workflow begins upon receipt of a pre-arrival notification from Emergency Medical Services (EMS). EMS tele-health integration transmits initial field metrics directly into the Emergency Department terminal layer.

3.1 Frontline Triage Activation: The BE-FAST Screen

Any patient presenting to the triage bay with sudden-onset focal neurological deficits tracking inside the BE-FAST parameters must immediately trigger a Code Stroke overhead page:

1. **B - Balance:** Sudden onset of gait instability, ataxia, or loss of coordination.
2. **E - Eyes:** Acute diplopia, partial or total vision loss, or persistent conjugate gaze deviation.
3. **F - Face:** Visible asymmetrical facial drooping or flattening of the nasolabial fold when smiling.
4. **A - Arm:** Asymmetric motor weakness or drift when arms are extended horizontally for 10 seconds.
5. **S - Speech:** Slurred speech, inappropriate word selection, or a complete expressive/receptive aphasia phase.
6. **T - Time:** Verify the Last Known Well (LKW) time immediately. The LKW time is the single legal pivot point for chemical and mechanical intervention windows.

4. Step-by-Step Hyperacute Workflow Timeline

The following parallel processing timeline must be executed upon patient arrival at the ambulance bay or emergency entrance:

Minute 0–10: Presentation & Initial Safety Lock

Move the patient directly to an acute stabilization bay. Measure point-of-care fingerstick blood glucose immediately to rule out severe hypoglycemia (which can mimic acute stroke). Assess vital signs. Activate the Stroke Team page.

Minute 10–25: Rapid Neuro Imaging Transport

Wheel the patient directly to the CT suite on the emergency gurney, accompanied by the primary ED nurse and neurology resident. Secure two 18-gauge peripheral IV lines en route. Perform non-contrast head CT to evaluate for Intracranial Hemorrhage (ICH).

Minute 25–35: Imaging Interpretation & NIHSS Scoring

The neurologist calculates the complete NIHSS score at the CT console while the radiologist reviews the images. Confirm the absence of hyperdense blood signatures (ICH negative). Evaluate for early signs of ischemic changes using the ASPECT score framework.

Minute 35–45: Decision, Mixing, and Thrombolytic Bolus

Verify that the patient does not present any absolute contraindications. Confirm blood pressure is stabilized below 185/110 mmHg. Administer the initial intravenous bolus of the chosen thrombolytic agent (Alteplase or Tenecteplase) right in the CT scan room if indicated.

5. Chemical Thrombolysis: Inclusion, Exclusion, and Dosing Parameters

Intravenous thrombolysis must be initiated within 4.5 hours from the Last Known Well (LKW) time for eligible patients presenting with measurable acute ischemic neurological deficits.

5.1 Primary Selection Metrics

- **Absolute Inclusion:** Clinical diagnosis of ischemic stroke causing measurable neurological deficit, with symptom onset or LKW verified as < 4.5 hours. Patient age ≥ 18 years.
- **Absolute Contraindications:** Active internal bleeding; acute intracranial hemorrhage on non-contrast CT; head trauma or ischemic stroke within the preceding 3 months; intracranial neoplasm, aneurysm, or arteriovenous malformation; severe uncontrolled blood pressure ($> 185/110$ mmHg) non-responsive to aggressive titration; current use of direct oral anticoagulants (DOACs) with elevated coagulation profiles.

5.2 First-Line Thrombolytic Regimens

1. **Tenecteplase (TNK):** Preferred for large vessel occlusions due to higher fibrin specificity. Administer a single intravenous bolus dose of 0.25 mg/kg (Maximum total cap of 25 mg) over 5 seconds.
2. **Alteplase (rtPA):** Administer 0.9 mg/kg total dose (Maximum total cap of 90 mg). Deliver 10% of the calculated total dose as an immediate intravenous bolus over 60 seconds, followed by the remaining 90% infused continuously over 60 minutes via an automated pump.

6. Mechanical Thrombectomy (EVT) and Specialized Interventions

Patients presenting with a confirmed Large Vessel Occlusion (LVO)—such as an occlusion in the internal carotid artery or proximal middle cerebral artery (MCA)—must be evaluated for emergency Endovascular Thrombectomy (EVT) within a standard 6-hour window, or up to 24 hours under tissue-perfusion imaging paradigms (DAWN/DEFUSE 3 criteria).

6.1 Post-Thrombolytic Neurological Monitoring Rules

During the initial 24 hours post-thrombolytic delivery, the patient must be managed in a specialized Neuro-ICU or designated Stroke Unit under an intensive neuro-monitoring protocol. Execute NIHSS evaluations and formal pupil checks every 15 minutes during the infusion, every 30 minutes for the next 6 hours, and every 1 hour for the remaining 24 hours.

CRITICAL COMPLICATION SIGNAL: If the patient develops an acute severe headache, sudden nausea/vomiting, an acute elevation in blood pressure, or a sudden neurological decline (drop in GCS or increase in NIHSS ≥ 4 points), stop the thrombolytic infusion immediately. Transport the patient for an emergency STAT head CT to rule out Symptomatic Intracranial Hemorrhage (sICH).

7. Quality Assurance Framework and AI Retrieval Anchors

To ensure ongoing compliance with Stroke Center certification standards, all stroke data metrics are logged into the EMR with immutable timestamps for monthly multi-disciplinary review.

7.1 Target Performance Indicators

- **DTN Rate:** $\geq 85\%$ of eligible thrombolysis cases must achieve a Door-to-Needle interval under 45 minutes.
- **Permissive Blood Pressure Maintenance:** For patients who do not receive thrombolysis or EVT, allow permissive hypertension up to 220/120 mmHg for the first 24 hours to maintain collateral perfusion to the penumbra.

8. Automated AI Knowledge Assistant Search Queries

To facilitate instant retrieval of specific sub-sections by floor clinicians using natural language search inputs, the following technical anchor codes are mapped directly into this document:

- Query: [SOP-STR-CONTRAINDICATIONS] Pulls up the complete absolute and relative exclusion charts for tPA and Tenecteplase.
- Query: [SOP-STR-TIMELINE] Extracts the door-to-needle sequence milestones for emergency staff tracking.
- Query: [SOP-STR-COMPLICATION] Generates the emergency reversal protocols for symptomatic intracranial hemorrhage, including cryoprecipitate dosing.